HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and debugged **Linux kernel** modules for **DMA** operations, enabling 32-bit drivers to access 64-bit virtual memory spaces utilizing **Intel and AMD IOMMU controllers**.
- Programmed **networking chips** to perform latency analysis and maintain accurate event counts, processing billions of events per second for network telemetry.
- Implemented **Python scripts** to bootstrap new hardware bring-up, automating initial configuration and diagnostic checks on newly developed network devices.
- Built **software agents** to mediate and translate requirements between switching ASICs, custom FPGAs, media adapters, and PCIe bridges, ensuring seamless hardware-software interaction.
- Wrote **C/C++ code** for fault-tolerant, concurrent routing protocol agents (**BGP, OSPF, ISIS**) within an embedded Linux environment, ensuring high availability for data center networks.
- Developed **configuration management services** using **Go and Python** to support standardized switch configuration via **YANG, OpenConfig, NETCONF, and RESTCONF** over gRPC, integrating with tools like git and Jenkins.
- Containerized network agent services using **Docker** and deployed to a **Kubernetes** cluster (EKS), configuring basic resource limits for predictable rollout and scaling.

SKILLS

Languages – C, C++, Java, Python, Go

Networking – Linux Kernel, DMA, IOMMU, ASICs, FPGAs, PCIe, SPI, SMBus, I2C, LLDP, 802.1x, BFD, BGP, OSPF, ISIS

Cloud & Containerization – AWS (EC2, EKS, CloudWatch), Docker, Kubernetes

Configuration & Protocols – YANG, OpenConfig, NETCONF, RESTCONF, gRPC, SSH, SSL

Development Tools & Methodologies – Git, Gerrit, Jenkins, Trello, Agile, Scrum

Embedded Systems – Linux Embedded Systems, Hardware/Software Boundary

Data Handling – High-resolution timers (sub-100ns accuracy), Event counting, Latency analysis

PROJECTS

High-Performance Network Telemetry System

- Developed a **Python agent** to interface with ASIC registers over **SPI and PCIe**, collecting near-real-time packet flow data.
- Implemented low-level **C code** to process and aggregate hardware event counters with sub-100ns timer accuracy, enabling detailed latency analysis.
- Engineered data structures for efficient storage and retrieval of network statistics, supporting aggregation of billions of events per second.

Automated Network Device Configuration Service

- Built a **Go service** for managing network device configurations using **YANG models** and **NETCONF**, integrating with a **Git** repository for version control.
- Developed **Python scripts** to automate the deployment of configurations to network devices via **SSH and gRPC**, reducing manual configuration time by approximately 40%.
- Integrated the service with **Jenkins** for CI/CD pipelines, enabling automated testing and deployment of configuration changes.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)